

PIN-0005 · LIVING WHITEPAPER · 2026-09-04

P-PAS+ Adaptive Scheduling With Persistent Stillpoint Restore

candidate gates passed

40/40 repeated workloads

validated active profile

GOLDEN failure under retest

Executive summary

PROBLEM

Large cold prefills interfere with active decoding under scheduler pressure.

STRONGEST RESULT

+4.90% decode throughput and -4.52% decode wall time in the target paired scenario.

CURRENT STATUS

Validated active profile; unchanged GOLDEN failure is under focused retest.

CONTRIBUTION

P-PAS+ ported to vLLM 0.28 without removing LMCACHE, MTP-3 or long-context support.

VALIDATION

All candidate gates and 40/40 repeated candidate workloads passed.

KEY LIMITATION

The paired gain is a two-repetition median for one workload shape, not a universal speed claim.

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Abstract. We ported prefill-pressure adaptive scheduling to a production-like Qwen3.8-27B serving profile while preserving persistent LMCACHE restore, MTP-3, FlashInfer XQA, FP8 KV cache, 131k context and the OpenAI-compatible capability surface. P-PAS+ passed every candidate correctness and restore gate and completed 40/40 repeated

workloads. In a preliminary paired canary, it improved decode throughput by 4.90% in the intended one-large-prefill pressure case and was effectively neutral in the dual-prefill case.

Prior work and thanks. P-PAS and P-PAS+ were developed by **Timo Sämann**. We thank him for publishing the implementation and research openly. Original sources: [P-PAS+ GitHub repository](#) and [P-PAS: Prefill-Pressure Adaptive Scheduling for Long-Context LLM Serving](#). Stillpoint's contribution is an independent adaptation and validation on a different model, vLLM version, GPU class and cache architecture; it is not the original P-PAS+ work.

1. Problem statement

A long cold prefill can monopolize the scheduler token budget while active requests are decoding. Persistent Stillpoint restore eliminates prefill work for exact cached prefixes, but unavoidable cold or partially cached prefills still occur. P-PAS+ complements LMCache: restore avoids work when possible; adaptive scheduling limits interference when the work cannot be avoided.

2. Research questions and hypotheses

1. **RQ1:** Can P-PAS+ be ported to vLLM 0.28 and Qwen3.8 without violating the LMCache hybrid-alignment scheduler invariant?
2. **RQ2:** Does the candidate preserve deterministic output, long-context retrieval, LMCache restart restore, MTP-3, API compatibility, tool calling, reasoning and image processing?
3. **RQ3:** Does adaptive pressure capping protect active decode work when one very large cold prefill overlaps multiple decodes?
4. **RQ4:** Does the candidate avoid material regression when two medium-long prefills overlap two decodes?

H1: the candidate passes every declared feature-equivalence and restore gate. **H2:** decode wall time and throughput improve in the one-large-prefill pressure case. **H3:** the dual-prefill case remains within ordinary run variance. Any correctness failure, cache corruption, engine death, or material non-target regression rejects the candidate.

3. Related work

Sämann introduced Prefill-Pressure Adaptive Scheduling and the P-PAS+ heterogeneous-pressure extension for long-context serving [1, 2]. Stillpoint’s persistent restore work addresses a complementary layer: exact cached prefixes avoid prefill computation entirely, while P-PAS+ controls interference from prefills that remain unavoidable. The present work is an independent port and validation, not the original scheduling contribution.

4. Stillpoint adaptation

Upstream values could not be copied directly. Qwen3.8 with LMCache hybrid alignment uses a unified block size of 1600 and requires the configured `max_num_batched_tokens` to remain in `[1600, 3200)`. We retained the validated configured maximum and applied only a runtime pressure cap:

- `B_max = 3199`
- `B_cap = 2048`
- `large_prefill_threshold = 65536` remaining tokens
- Pressure rule A: at least two active prefills and one running decode
- Pressure rule B: one large running prefill and at least two running decodes

The configured maximum stays unchanged so MTP draft-slot accounting and LMCache input-budget validation retain their established behavior.

5. System under test and lab setup

Model	Runtime
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Qwen3.8-27B W4A16 AWQ

vLLM 0.28.0 · tensor parallel 2

GPU

2 × NVIDIA RTX PRO 2000 Blackwell 16 GB · SM120

Attention

FlashInfer XQA · PIECEWISE CUDA graphs

KV cache

FP8 · 160,799-token GPU capacity

Speculative decoding

MTP-3

Persistent cache

LMCache multiprocess · disk-backed L2

Hybrid invariant

Mamba align · unified block size 1600

6. Test suite and experimental methodology

The candidate had to preserve the complete production feature set, not merely start or improve one synthetic trace.

- Deterministic generation: 20 identical answers and five identical 200-token generations
- Buried-key retrieval at 13k, 30k, 60k, 66.7k, 112k and 130k
- LMCache store and restore from 5k through 112k, including restart recovery
- Model discovery, non-streaming completion and SSE streaming
- German output, reasoning parser, native tool calling and deterministic coding
- Real image processing path and eight-request queue submission
- Repeated short, agent, long-context, concurrent, heterogeneous and mixed-pressure workloads

7. Results: paired GOLDEN canary

Two identical repetitions were run per scenario. These are preliminary medians, not universal performance claims.

SCENARIO / METRIC	GOLDEN	P-PAS+	CHANGE
1 × 100k prefill + 3 decodes · decode wall	456.19 s	435.59 s	4.52% better

SCENARIO / METRIC	GOLDEN	P-PAS+	CHANGE
Same · decode throughput	1.751 tok/s	1.837 tok/s	4.90% better
Same · makespan	457.07 s	435.88 s	4.64% better
2 × 56k prefills + 2 decodes · decode wall	134.13 s	134.44 s	0.23% worse
Same · decode throughput	5.957 tok/s	5.944 tok/s	0.23% worse
Same · makespan	134.12 s	134.65 s	0.40% worse

8. Results: clean candidate validation

Each workload below completed five repetitions without a candidate request error.

WORKLOAD	MEDIAN WALL	MEDIAN RATE	RESULT
Short decode · 512 output	9.522 s	54.491 tok/s	5/5
8k agent · 512 output	17.698 s	49.501 tok/s	5/5
111k prompt · 256 output	142.437 s	43.521 tok/s	5/5
4 concurrent short	12.519 s	163.597 aggregate tok/s	5/5
4 concurrent 111k	569.211 s	0.899 aggregate tok/s	5/5
Heterogeneous 16k–130k	304.698 s	2.310 aggregate tok/s	5/5
1 large prefill + 3 decodes	474.043 s	1.730 decode tok/s	5/5
2 prefills + 2 decodes	140.252 s	5.698 decode tok/s	5/5

MTP acceptance was 59.15% in the steady suite and 61.75% under mixed pressure.

9. Failure analysis and current interpretation

The extended GOLDEN arm started successfully, passed its real health check and completed five short-decode repetitions. During the second 8k-agent repetition, both workers stopped on

CUDA OOM in the MTP proposal path and vLLM raised [EngineDeadError](#). P-PAS+ completed the full candidate sequence.

This proves that the P-PAS+ profile is operational and stable under the tested sequence. It does not by itself prove that P-PAS+ universally uses less memory, prevents OOM, or is faster for every workload. The failed GOLDEN case is being rerun unchanged; this paper will be updated with the outcome.

10. Threats to validity, caveats and non-claims

- The paired A/B figures are medians from two repetitions per scenario and do not establish production-wide performance.
- The strongest improvement was workload-specific; the dual-prefill result was effectively neutral and slightly favored GOLDEN numerically.
- The GOLDEN OOM does not prove that P-PAS+ uses less memory, prevents OOM, or caused the difference.
- Results apply to the declared Qwen3.8, vLLM 0.28, LMCache, MTP-3, FlashInfer and dual-SM120 setup.
- Historical champion numbers were not substituted for the incomplete contemporaneous extended control arm.

11. Reproducibility and artifact availability

- Upstream implementation: [TimoSaemann/ppas-vllm](#)
- Upstream paper: [arXiv:2608.15171](#)
- Stillpoint adaptation: vLLM 0.28 scheduler port with a switchable isolated runtime profile
- Evidence set: raw JSON gates, benchmark results, service state and OOM journal, all SHA-256 verified after transfer

The engineering repository retains the exact patch, pinned upstream snapshot, raw evidence and SHA-256 manifests. Public artifact links will be added when that repository's disclosure

review is complete.

12. Conclusion

P-PAS+ is a validated active Stillpoint profile: it preserves the full production capability set and completed the entire repeated candidate campaign. The preliminary paired evidence supports the intended scheduling benefit when one very large prefill competes with active decoding, while showing no meaningful benefit in the dual-prefill case. The focused unchanged GOLDEN retry determines whether the extended-run OOM is reproducible; it does not change the candidate's independently passed gates.

13. Acknowledgements

We thank Timo Sämman for publishing P-PAS/P-PAS+ and its implementation openly, enabling independent adaptation and validation. We also acknowledge the vLLM, LMCache and FlashInfer open-source communities whose software forms the evaluated stack.

14. References

1. T. Sämman, "P-PAS: Prefill-Pressure Adaptive Scheduling for Long-Context LLM Serving," arXiv:2608.15171, 2026. <https://arxiv.org/abs/2608.15171>.
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3. vLLM Project, "vLLM: Easy, Fast, and Cheap LLM Serving with PagedAttention," software version 0.28.0. [GitHub repository](#). Accessed 2026-09-04.
4. LMCache Project, "LMCache," persistent KV-cache software. [GitHub repository](#). Accessed 2026-09-04.
5. FlashInfer Project, "FlashInfer," attention kernels for LLM serving. [GitHub repository](#). Accessed 2026-09-04.

15. Revision and run history

1. **2026-09-04 · Preliminary paired canary:** target case improved by approximately 5%; dual-prefill case neutral.
2. **2026-09-04 · Clean candidate validation:** all gates and 40/40 repeated candidate workloads passed.
3. **2026-09-04 · Extended GOLDEN arm:** CUDA OOM during the second 8k-agent repetition; focused unchanged retry in progress.

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